

ScreenPlink

Turn your monitor into a laser dry-fire range

ScreenPlink

User Guide

Version 1.0

October 2026

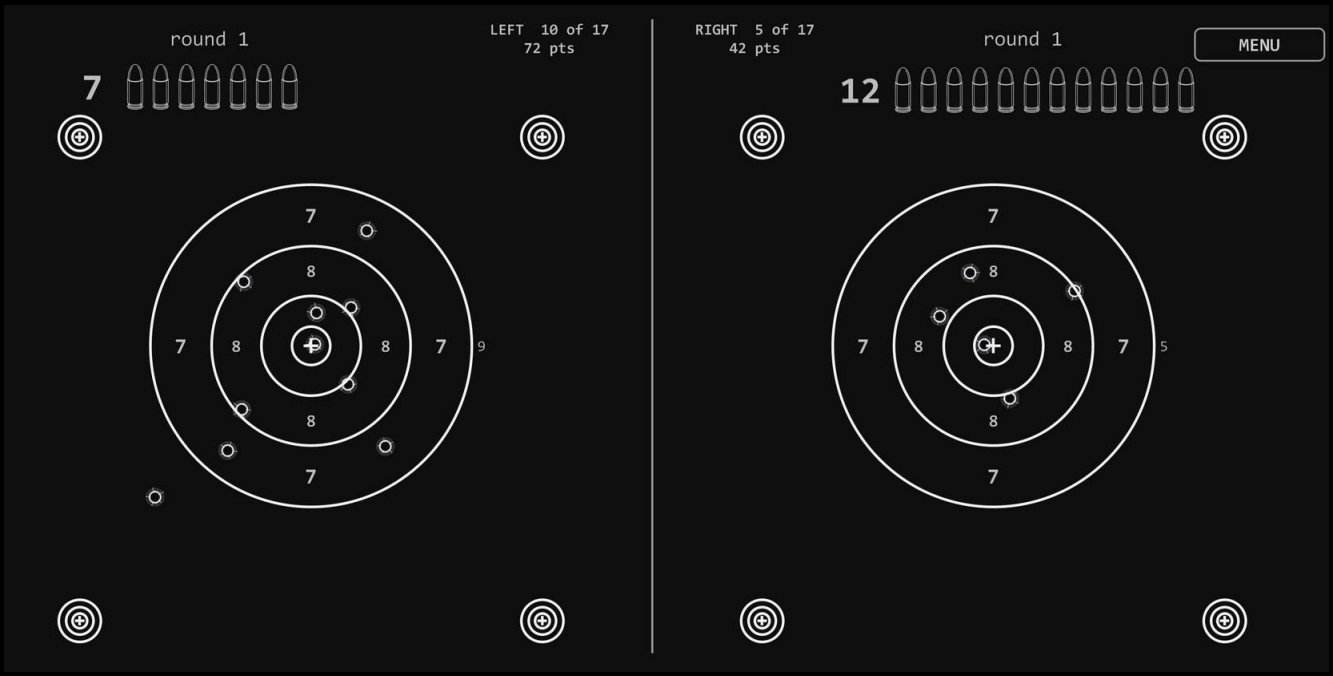
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WHAT SCREENPLINK IS

ScreenPlink turns a monitor you already own into a shooting range. The screen shows targets. An ordinary USB webcam watches the screen. When you fire a laser trainer pistol at it, the app works out which pixel the dot landed on, draws a bullet hole there and plays a shot.



Static Targets, in two-player mode. Each side has its own magazine, its own round counter and its own score.

There is no projector, no special screen and no sensor bar. The camera does not have to sit in front of the monitor: off to one side at an angle is normal and fully supported, because the app measures the mapping from the camera's view to the screen rather than assuming one.

Why everything is grey

Nothing in this app is drawn in colour, and nothing is drawn brightly. That is not a style decision, it is the only way the thing works.

The camera is hunting for a red dot. It scores every pixel by how much redder it is than it is green or blue, so a red graphic anywhere on screen builds a target that shoots itself. Early versions did exactly that: an amber warning panel and a coloured status bar fired dozens of phantom shots in a single session.

Brightness matters for the opposite reason. A laser dot landing on a lit grey area loses that grey from its score, one unit for one unit, because the red is already saturated and the green and blue underneath it are not. A target filled with bright grey is a target that eats the shots aimed at it.

So every target, every number and every box is neutral grey on black, and the brightest grey the app will draw is taken from the darkest of the target drawings on disk. Add a darker drawing and the whole interface dims to match it.

Sizes, not pixels

Nothing on screen is sized in pixels. Every target, every piece of text and every box is a fraction of the screen, so the same drill plays the same way on a 24-inch monitor as on a 43-inch one, and a 4K screen is not four times harder than a 1080p one for no reason.

Size is not the constraint. **The surface is**, and it is the one requirement in the app that nothing else can make up for, so it has the next chapter to itself.

THE SCREEN IT NEEDS

ScreenPlink needs a matte, anti-glare computer monitor. This is the one hardware requirement in the app that cannot be worked around by moving anything, changing a setting or buying a better camera, so it is worth settling before you set anything up.

Glossy, semi-gloss and “glass-look” anti-reflective screens are not supported. That includes most modern televisions. On those surfaces a laser hit fired from an angle is frequently undetectable.

Display	Works
Matte or anti-glare computer monitor	Yes
Glossy or semi-gloss monitor	No
Glossy television – most LCD, QLED and OLED sets	No
Projector and screen	Not in version 1

A projector is the one entry on that list that is not about gloss. A projection screen is diffuse and would scatter the dot perfectly well. The obstacle is the palette: this app draws light grey on black, and a projector makes black by projecting nothing, so the dark parts of the picture are simply whatever the room does to a white screen. Making it work means inverting the whole scheme – dark targets on a lit field – together with every brightness rule derived from the present one. That is a different app rather than a setting, and it is not in this version.

Why the surface decides it

The app finds your shot by watching the screen and looking for the laser spot. It can only see that spot if the screen scatters some of the beam back toward the camera.

A **matte** screen has a micro-textured coating that throws incoming light in every direction at once. The spot is visible from almost anywhere, which is the whole reason an ordinary webcam parked off to one side works at all.

A **glossy** screen behaves like a mirror. Shoot it at an angle and most of the beam leaves at the matching angle on the other side, away from the camera. An anti-reflection coating weakens what little comes back, and the panel’s own layers absorb a good part of the remainder.

So a glossy screen often registers hits fired square on and misses most of the ones fired from an angle. **Moving the camera does not fix it.** The mirror direction depends on where on the screen the shot landed, so it changes with every shot, and no fixed camera position can follow it.

What has actually been tried

Display	Finish	Result
Dell UltraSharp U4320Q, 43-inch 4K	Matte	Hits visible at every angle tried
Samsung 75-inch QLED, 2019 Q7 series	Glossy, anti-reflective	Sometimes visible head-on, mostly invisible at an angle

The Dell is the screen this app was built and measured on. Everything in this guide – every threshold, every timing, every number – came off it.

Why no other app asks this of you

The requirement reads like a limitation until you look at what everything else in this category asks for instead. Checked on 22 October 2026, against the vendors' own published documentation.

Laser dry-fire products divide into two kinds, and **neither of them shoots at a screen**. Most watch a **printed paper target**: the camera is pointed at the target on the wall, and a television, if one is involved at all, is only there to show you your hits from across the room. The rest **project** onto a screen or a white wall, and the surface they specify is a matte one – one system calls for "a white projection screen (1.0 gain max, no silver screen) or flat, non-glossy, white wall", which is the same physics as this chapter arrived at from the other direction.

Two products address using a flat screen as the target. **Both of them tell you to cover it first.**

One recommends against the idea altogether – *"we recommend using a projector instead of a flat screen (TV, monitor, LED, LCD, etc)"* – and its description of what goes wrong is this app's symptom, written by someone who has never seen it:

"you see a bright laser (red or green) dot on the screen, but from the camera the same laser dot is not even visible"

Its remedy is to stick a transparent film made for laser pointers onto the screen: one *"transparent enough to see through but frosted enough to reflect the laser"*. That is, to stop it being glossy.

The other is the largest system in the category, and its own setup guide for televisions is blunter still:

"Use with TV or monitor will require our filter material that is taped to the screen in order to track the laser and help control reflections."

"Some TVs will only reflect the laser at an angle incident to the angle the laser enters the screen. Therefore the camera will not be able to see the laser and track it."

"Some TVs have a protective layer that is not directly on the screen surface… There is no way to fix this unfortunately and this means it will not work."

The material they supply for the job is a vinyl tablecloth. The middle quotation is this chapter's physics, in the words of the people who sell the market-leading system – and their answer to it is to tape something over the panel.

Nothing goes on your screen here. The requirement is instead that the panel be matte to begin with, which is the same fact approached from the other side: a frosted film and a matte coating do the same job, and one of them is already on the monitor.

AND THE DISAGREEMENT, WHICH IS THE INTERESTING PART

That same manual blames **anti-glare coatings** for losing the dot. This app measures the opposite: anti-glare is the finish that works. Both can be true, and what reconciles them is where the camera stands. That manual also advises placing the camera **directly in front of the screen**.

Square on, a glossy panel's mirror reflection comes straight back toward the camera, so gloss helps and a matte coating spreads the same light away. Off to one side, the mirror reflection leaves on the far side of the screen and only the diffuse scatter ever reaches the lens – so matte becomes the only thing that works and gloss fails. The glossy television in the table above behaved exactly that way: sometimes visible head-on, mostly invisible at an angle.

So the matte requirement is the price of the camera being off to one side, and that is the trade this app makes on purpose. It is what lets you shoot at a monitor you already own, with nothing standing between you and the screen, no projector to cast your own shadow across, and no paper to print and replace.

How to tell what you have

Look at a reflection. Switch the display off and stand so that a lamp or a window is behind you.

- A sharp, mirror-like reflection means glossy, and it will not work properly.
- A blurred, diffuse glow means matte, and it will.

Read the specification. "Anti-glare", "matte" or "3H anti-glare hard coating" is what you want. "Glossy", "glare type" or "anti-reflection glass" is what you do not.

Then let the app look at it. Press **C** and it runs its self check: it draws a dozen red dots at known places on your screen at the brightness of real shots, photographs each one, and reports how many it found and how far out it placed them.

Be careful how you read that number, because it answers a narrower question than it appears to. A screen *emits* its own test dot, while your laser *reflects* off the panel's coating, and the two fall off across a screen quite differently – measured here, the app's own dot held roughly steady from one side of the screen to the other while a real laser dropped by half. So:

- the **position error** it reports is real, because it is measured through the same camera-to-screen mapping your shots are;
- the **detection rate** it reports is not your pistol's. It proves the app can see dots on your screen and place them accurately. It cannot tell you what fraction of your trigger pulls will register.

The honest test of a glossy screen is still a magazine of real shots fired from where you intend to stand.

Size, and getting the dot big enough

Any size of matte screen is fine, and nothing in the app is laid out in pixels (see the previous chapter), so a drill plays the same on a 24-inch monitor as on a 43-inch one.

What changes with size is how many camera pixels the laser spot lands on. The bigger the screen, the smaller a share of the camera's view each shot occupies. On a large display, either use a camera with more resolution or move it

closer, so that the screen fills the frame with only a modest margin around it. A spot spread across four camera pixels is near the limit of what can be told from sensor noise; the same spot across twenty is unmistakable.

Two settings matter for the same reason, and the app handles both by itself: the camera's exposure and gain are driven down and locked, because automatic exposure brightens the whole picture and washes the spot out, and the targets are drawn dark, because any lit grey underneath the dot is subtracted from it, one unit for one unit.

QUICK START

What you need

- **Windows 10 or 11**, and a **matte, anti-glare monitor**. Not a glossy screen and not a television – see the previous chapter, which is about nothing else.
- **A USB webcam that delivers sixty frames a second.** A laser pulse lasts about a tenth of a second, so at sixty frames it lands in about six of them and the app needs two to agree before it will call a shot. At thirty it lands in three, which leaves almost no margin, and a camera that quietly drops to ten – which several do, without saying so – puts the whole pulse in a single frame, where nothing works. Check the measured rate rather than the box: **Doctor.cmd** times real frame delivery and says how many frames a pulse would land in.
- **A laser trainer cartridge or pistol** that flashes a red dot when the trigger breaks.

Unzip the folder anywhere and keep it together. Nothing is installed and nothing is written to the registry.

Five steps

1. Aim the camera at the screen. It needs the whole display in view with a margin around it. It may sit to the left or the right, above or below, at an angle. It must not be able to see out of a window if you can help it.

2. Start it. Double-click **Play.cmd**. The app opens full screen on the monitor the camera is watching.

3. Do not fire while it measures. For the first half minute it flashes the screen, draws small markers on it and fires test dots at itself. The screen says **HOLD FIRE** while that is happening. A real shot fired during the noise measurement makes *that shot* the noise floor, and then the detection bar sits above every genuine hit for the rest of the session, which feels exactly like the app working for a few shots and then stopping.

4. Shoot a tile on the menu. When the measurements are done the Main Menu appears. Shoot the square you want, click it, or press the key printed on it.

5. Shoot. Every shot that registers draws a hole and plays a bang. You have seventeen rounds. When the magazine is empty the round ends and the screen shows what you did with it.

To leave, press **Escape**, or shoot the **Exit** tile twice.

Three things that look like faults and are not

The app seems to fire shots you did not fire. Something coloured is on the monitor the camera watches: another window, a notification, a status panel from some other program. Move it off that screen.

The mouse pointer disappears. It is hidden on purpose while you shoot, and comes back the moment you move the mouse. A large bright cursor sitting on the panel is a bright object on the screen the camera is reading: measured here, it took the noise floor from 3 to 17 on its own and blocked a launch.

The signal is weak, so you reach for the light switch. Do not. A dark room is the best condition this app can have, because darkness is what lets it use a longer exposure, and the laser's brightness scales with exposure time. If shots are being missed, press **X** to measure the exposure against the room as it is now, or move the camera closer to the screen.

If it will not behave

Close the app and run **Doctor.cmd**. It tests every camera on the machine, writes its findings to *doctor-report.txt* and shows them. It will tell you whether the camera itself is fit to shoot with, which is the first question worth answering.

One safety line

This is a dry-fire trainer. It assumes a laser cartridge or a dedicated laser pistol and no ammunition anywhere in the room. The app has no way of knowing what is in your hand, so that part is yours.

WHAT THE APP DOES BEFORE IT LETS YOU SHOOT

The half minute between launching and the menu is the app measuring itself. It matters because the chain from a trigger pull to a bullet hole is a dozen stages in series, and a weak link anywhere in it looks identical to every other weak link from the firing line.

In order, it:

- **finds the screen** by flashing the monitor white and then black and subtracting one from the other;
- **measures the exposure** by drawing its own red dot and photographing it at several camera settings, then picking the one where the dot stands furthest clear of the background;
- **measures the noise floor** – the brightest red thing in view when nothing is being fired – and sets the detection bar from it, rather than from a guess;
- **confirms the camera-to-screen mapping**, by drawing two markers at known places and finding them. If they do not confirm, it measures the whole mapping again from eighteen markers;
- **fires a dozen test dots at itself** at the brightness of real shots, and counts how many came back.

Then it says what it found. A green verdict means it detected its own shots and is ready. A problem names itself and names the one thing to do about it.

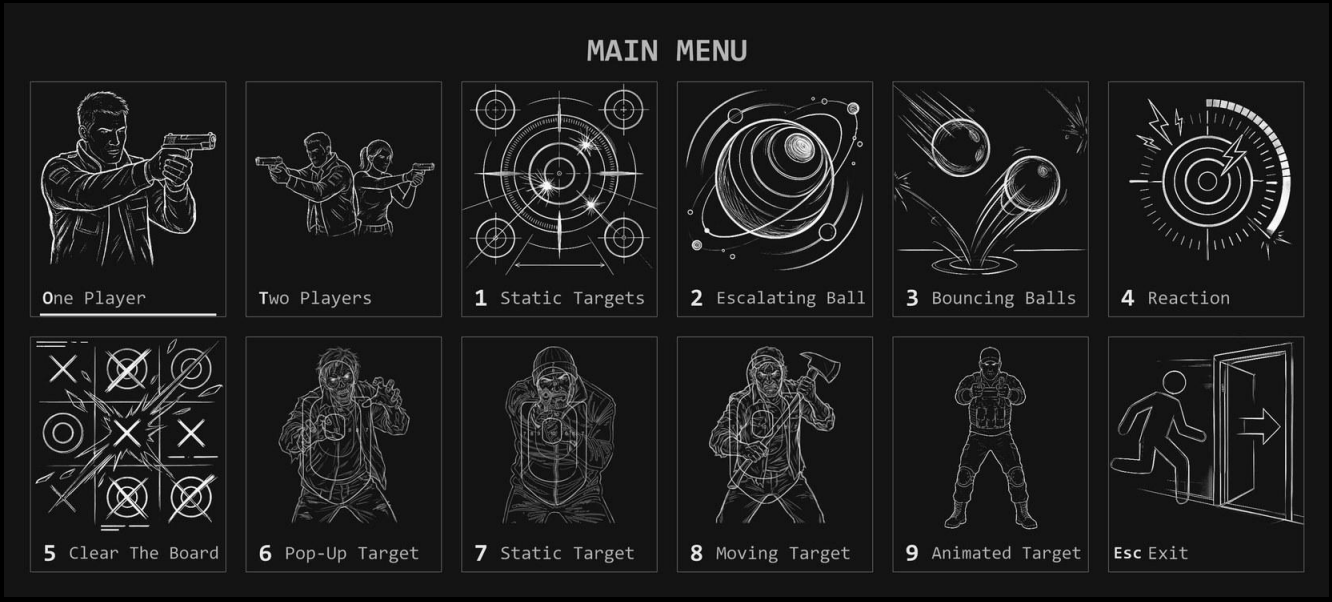
This is the part to leave alone. Every step photographs the screen, so a shot fired into it is measured as part of the room.

Afterwards you can re-run any of it by hand, and it is worth knowing which key does what:

Key	What it does	When you want it
C	Self-calibrate	After moving the camera
A	Self-check	“Is it seeing my shots?”
Q	Show the last verdict again	It cleared before you read it
G	Calibrate by shooting a grid	The mapping is wrong and the markers will not fix it
X	Measure the exposure	The light in the room has changed
F	Measure the noise floor	Something bright was on screen during setup

The first launch on a new machine takes longer than every later one, because it has nothing remembered to check against.

THE MENU, AND MOVING BETWEEN DRILLS



The Main Menu. Two player-count tiles, nine drills, and Exit.

Press **M** for the menu at any time. Every tile can be shot, clicked with the mouse, or started by pressing the key printed in its corner.

How a choice is made

A shot at a tile does not switch the drill on the spot. It records the choice, and the app acts on it a fraction of a second later, on its next tick. There is one route for switching drills, and it is the route that announces the new drill aloud, writes it to the log and resets the score. A second route that skipped those three is the kind of thing that quietly rots.

The first choice in a turn wins. Two pistols can land two shots in the same camera frame, and a later one does not overwrite an earlier one.

Nothing about the tiles is typed into the menu. They are whatever drills the app has registered, in key order, with their own names. A drill added in a later version appears here by itself, correctly labelled, with no risk of the menu describing a drill that no longer exists – which the old hand-typed key legend did for three days.

Starting a new round

Press a drill's own key again to restart it: a fresh magazine, a cleared score, and the holes wiped. **N** does the same thing for whatever is running. When a round has finished, the **NEXT ROUND** box on the round screen starts the next one; it can be shot, clicked or started with N, and it is armed 0.8 seconds after it appears so that a shot still in flight cannot press it.

One player, two players

O is one player and **T** is two, and the first two tiles on the menu do the same thing. The choice applies from the next drill you start.

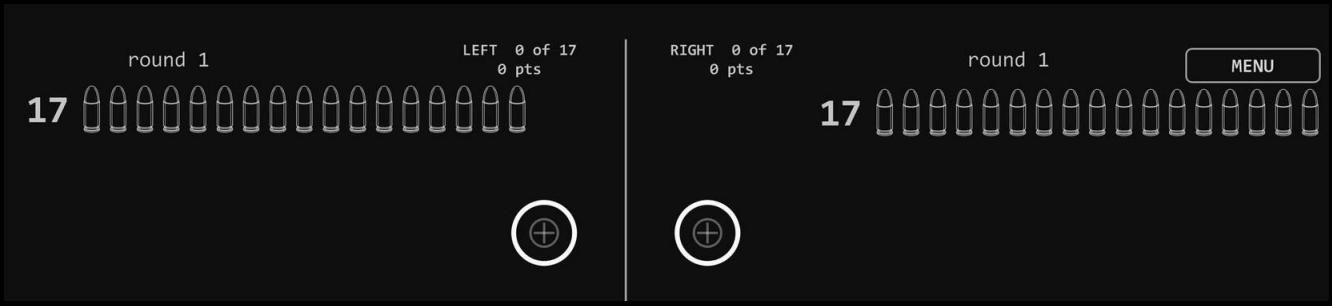
Exit

The **Exit** tile closes the app, and it is labelled Esc because Escape does the same thing. **It takes two hits.** The first arms it and the tile says so; the second closes the app, and the arming lapses after four seconds. A phantom detection closing the app in the middle of a string is the worst accident available here, which is why this one tile is the only thing in the app that asks twice.

Why the menu is not up while the app measures

Eleven lit tiles are a lot of light on a screen the camera is trying to read. The app therefore spends its measuring time on Static Targets – thin rings on black – and only brings the menu up when it has finished. Measured on the same machine minutes apart: with the menu up the noise floor read 3 and the test dots on the far side of the screen measured 13; on Static Targets the floor read 1 and the same dots measured 32.

THE MAGAZINE: SEVENTEEN ROUNDS



A full magazine on both sides, drawn one bullet per round. The numeral sits at the fixed end.

A round of ScreenPlink is one magazine. The default is **seventeen**, because that is what a CZ Shadow holds, and the point of the whole design is to train the awareness of the magazine you are actually carrying.

"This is not just the game. This is training, and people should understand the exact size of their mag and the exact number of bullets they have left in that mag."

The rules, and there are only four

Every shot the app registers spends one round, hit or miss. Nothing gives a round back.

The round ends when the magazine is empty. Not on a timer, not on a target count. One number decides both how many bullets are drawn and how long the round lasts, so the strip can never mean anything other than what is left.

The strip empties from the far end. The bullets nearest the anchor never move, so a glance finds the same bullets in the same place every time. The numeral at the fixed end is there because reading “12” from several metres away is faster than counting twelve outlines.

A pull on an empty magazine is a dry pull. It draws its hole and plays a bubble pop instead of the gunshot – the gunshot means a live round went off, and that was not one. It scores nothing and moves no hit rate. The strip tells you the gun is dry before the pull; the pop tells you after it.

The last hole

When the last round goes, the app waits about 1.2 seconds before putting the round screen up.

“I want to see the hole of my last shot for at least a second. As soon as I fire the number 17, that hole is supposed to be there, and you wait at least a second until you display the end-of-round screen.”

So the last hole is seen where it landed, against the target it struck, rather than being swept away by a results screen the instant it arrives.

Changing the size

The magazine is stored in *config\settings.local.json* as *magazine*, and may be anything from 1 to 50 – beyond that the strip stops being readable at a glance. There is no on-screen control for it yet. A change takes effect at the start of the next round rather than ending the one you are shooting.

THE SCORE, AND THE RECORD BOOK

At the end of every magazine the app shows two figures: a **score**, which is you measured against a perfect shooter on that same round, and an **efficiency**, which is points per second. The whole screen, single player, is this:

ROUND 2

17 shots
16 hits
1 miss
70 of 75 points
14.6 seconds

EFFICIENCY 4.79
= 70 points / 14.6 seconds

SCORE 256
AIM 93% SPEED 27%
BEST SCORE SO FAR 212
BEST EFFICIENCY SO FAR 3.99

[NEXT ROUND]

The tally at the top is measurement you can check yourself. *70 of 75 points* is your points beside the most a perfect shooter could have taken from the targets you were shown.

The screen never advances by itself. It waits for a shot at the **NEXT ROUND** box, a click on it, or **N**. The box is armed 0.8 seconds after it appears, so a shot still in flight when the magazine emptied cannot press it.

The score: you against a ghost

A *ghost* shoots the same round you just shot. It hits every target in the best zone available, and it fires as fast as the round allows – no faster than the app's own floor on shot cadence, and charged exactly the same unavoidable waiting the drill imposed on you, because waiting for a target to appear is not something either of you can shoot your way out of.

Both of you are then scored the way the round is already scored – points over the clock – and the comparison is the score:

SCORE = 1000 x aim x speed

aim = your points / the ghost's points how well you shot
speed = the ghost's seconds / your seconds how fast you shot

Both factors are shown, because the single number cannot say which you are. Across the rounds this was measured on, the median aim figure was 82 per cent and the median speed figure 21: careful shooting, taken slowly. One number would have reported a middling score and left you guessing which half to work on. *AIM 93% SPEED 27%* tells you to speed up and leave your aim alone.

A score of about **1000 means you matched a perfect shooter at this machine's own pace**. That is the intent of the scale, and it is reachable: on a drill with no waiting, a shooter firing at the app's cadence floor and hitting every best zone scores exactly 1000.

One rough edge, stated rather than left for you to find: on the drills where *dropping* a target starts a wait – a figure's fall, the soldier going down – the ghost drops more targets than you do and therefore waits more than you did, which can push a score past 1000. The highest seen here is 1146. It is a known defect in the comparison, not a reward for anything you did, and it is being fixed. Treat 1000 as "as good as this machine can be shot", not as a ceiling.

THE GHOST'S PACE, AND WHY IT IS FROZEN

The ghost fires one shot every **0.25 seconds**. That is not a guess at how fast a trigger finger moves; it is measured, and it is a property of *this app*. Across three thousand pairs of consecutive shots in real logs, the shortest gap the detector has ever accepted between two shots at the same place on the screen is 0.254 seconds – the length of the tracker's own refractory window.

That number is deliberately frozen. The score is directly proportional to it, so if it were ever quietly re-tuned, every record ever set would change meaning without anybody touching the record. Each stored score therefore carries the cadence it was computed against, and a record set against a different one is ignored rather than compared as though it still meant the same thing.

The efficiency: points per second

Points earned divided by seconds taken, and the sum is printed underneath so there is nothing to take on trust. The seconds are rounded to one decimal first and the division is done on the rounded figure, so the line below always equals the figure above it to the digit.

The clock runs from the instant the first round breaks to the instant the last one does, from the shots' own timestamps. Time spent settling before the first shot costs nothing; a slow string costs everything.

Nothing is averaged across rounds and no running mean is kept. One magazine, one figure, on its own screen.

WHY THERE IS NO PENALTY FOR A MISS

"When you pull up your gun and you have your mag and you need to do what you can with that mag, therefore you should be judged based on how you spend those 17 bullets, nothing else."

The magazine is the penalty. A round that misses is spent and cannot be recovered, so it has already cost its full value by earning nothing. Subtracting a further ten points would charge twice, and the size of the second charge would be invented. A shot that lands on a figure but outside every scoring ring is the same case: it earns nothing, it spends its round, and that is the whole of its cost.

What earns points

Drill	Points for a hit
Static Targets, and the figure drills	The ring it landed in: 10, 9, 8 or 7
Escalating Ball, Bouncing Balls	The level the ball had reached
Clear The Board	1 per circle
Reaction	10 for a quarter of a second, falling to 1 past a second and a half
Animated Target	10 for any hit; there are no rings printed on him to grade

On the figures, the rings that score are the rings printed in the drawing, and a hit inside the figure but outside all of them **scores nothing**. That is not an edge case: between 42 and 51 per cent of the ink in each figure – heads above the top ring, limbs, the weapon – lies outside every scoring ring. Such a shot leaves a dim grey mark so that it is visibly different both from a miss and from the app not having seen it. Note also that the artwork prints a 9 and a 7 on the head with no ring behind either: a hit there is on the figure, not in a scoring zone.

The record book

BEST SCORE SO FAR and *BEST EFFICIENCY SO FAR* are what this machine has seen. When a round beats one, that line becomes *NEW BEST SCORE RECORDED* or *NEW BEST EFFICIENCY RECORDED* in its place.

Records are kept **per drill, per magazine size and per player count** – the Moving Target with a magazine of seventeen in two-player mode is its own entry. Drills hand the ghost wildly different amounts of dead time, so a single global best would make most of them unbeatable for ever; and a best set on a magazine of five cannot be beaten on a magazine of seventeen. Each entry stores the ingredients as well as the result – the points, the seconds, the shot

count, the time it was set and whether the comparison was fully measured or partly assumed – so an old record can be re-derived rather than orphaned if the model changes.

Three things to know about it:

It says "best so far", not "personal best", and the wording is exact. There is no shooter identity anywhere in the app – nothing to log in to, nothing to select, nothing stored about who fired. The player count in the key is a *count*, not a name. Two people shooting the same machine share one record book.

There are no lifetime totals and no round history. Only the best per entry is kept. Every round's full detail – every shot, every reject, every figure – goes to the log in *logs*\ and nowhere else.

It lives in *configsettings.local.json*, under *bests*.** Deleting that one section resets every record and leaves your calibration alone, which is what you want: the calibration is the part that took real shots to learn.

A round is sound, or it is discarded

Sometimes the app sees shots nobody fired – something coloured appearing on the monitor it is watching will do it. A burst like that can fill a magazine in two seconds and score higher than any real round, and once that is in the record book it can never be legitimately beaten.

So a round the app cannot believe is thrown away whole, and says so:

ROUND 2

17 shots
16 hits
1 miss
70 of 75 points
2.7 seconds

EFFICIENCY --

ROUND DISCARDED
17 shots in 2.7 s is faster
than a pistol can fire.
No score, nothing saved.

The tally stays, because that is measurement you can verify for yourself. The judgement disappears: no score, no aim, no speed, no efficiency, nothing stored. There used to be a middle state – a score shown with a note saying it would not be kept – and it was rightly thrown out:

"How does it make sense to you to display the score and at the same time declare it invalid?"

The app never shows a figure it will not honour. The converse holds too: a score that appears on the screen is recorded if it beats the stored best. There is no third answer.

Only the rate of fire can discard a round – shots arriving faster than the tracker is able to accept them, which no trigger finger can do. Dim shots and a quiet microphone are measured and written to the log, and they never throw a round away: a dim median is the normal state of a real session, and discarding a good round for it would be worse than the fault it guards against.

When the app marks its own figure

If the app's own measurement was poor while you were shooting, the efficiency is starred and the reason is named:

*EFFICIENCY 3.10 **

** measurement was poor this round: something covered the screen for 6s*

The conditions are read from measurements the app already holds: something covering the screen, the camera appearing to have moved, a thin signal margin, the detection bar shifting mid-round, or the microphone hearing trigger pulls the camera never saw.

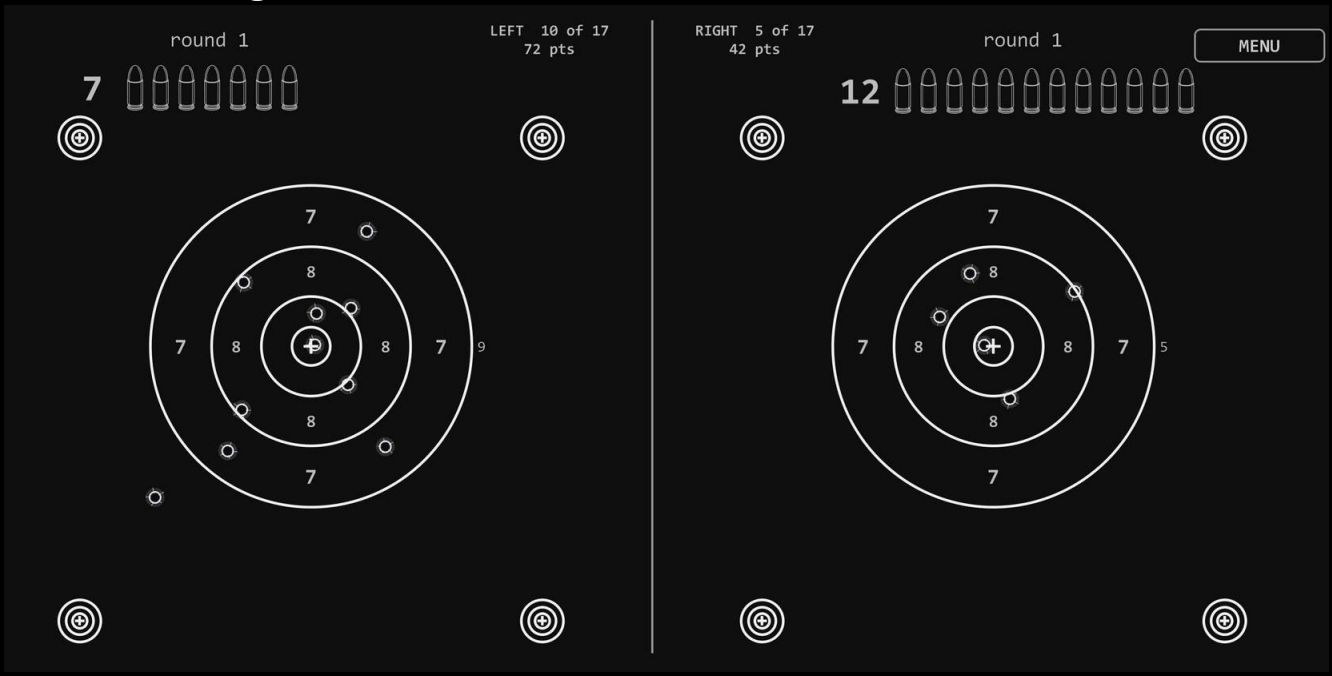
This exists because of a rule the app follows everywhere: **it never coaches you on its own blindness.** A figure depressed because the app could not see half your shots is "you are pulling left" with a number attached, and you would train away a flinch you do not have.

What this is not

This is not USPSA or IPSC scoring and it is not hit factor. The efficiency figure is the same shape of arithmetic answering a different question, by choice. Hit factor subtracts penalties before dividing and assumes a power factor and a rulebook edition; none of that is here, and nothing in this app should ever be described as if it were.

THE DRILLS: ONE TO FIVE

1 - Static Targets



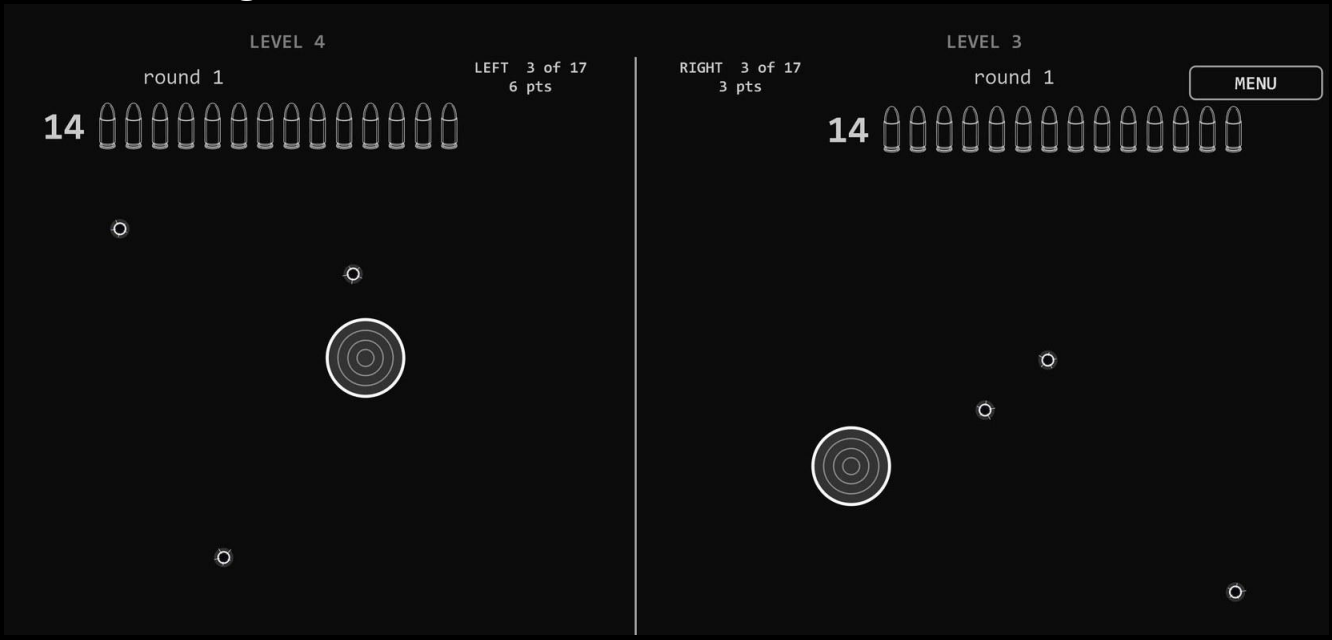
A big target in the middle, four small ones in the corners, and the rings that score.

A large ringed target in the middle and four small aim marks in the corners. Nothing moves, nothing escalates and nothing disappears: you shoot, holes appear, that is the drill. Each ring is worth one more than the ring outside it: the big target scores 7 to 10, and the smaller corner marks 7 to 9.

This is what the app lands on while it is measuring, and what key 1 returns to. It replaced a blank “free aim” screen, which was removed for the best of reasons: *“Is free aim just a blank screen without anything? It is useless to me.”*

A hit flashes a brief ring on the mark it struck, and each mark keeps a small tally of the hits it has taken, so the shape of a string can be read off the screen without turning on any readouts.

2 - Escalating Ball



One ball, and the level it has reached. Each side escalates on its own.

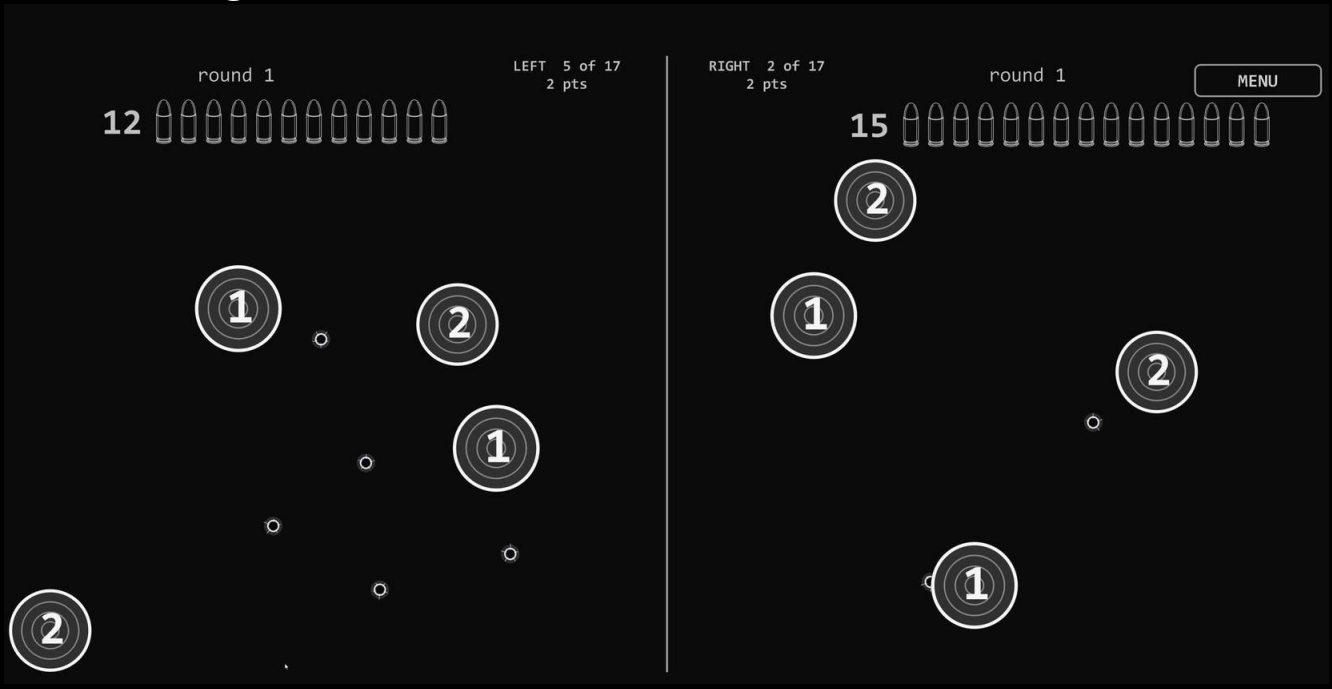
One ball bounces around. Every time you hit it, it comes back five per cent faster, up to level five and no further. Hitting a level-four ball is worth four points, so the score is a record of how far you got.

Three decisions worth knowing:

- **The cap is real.** It was uncapped at ten per cent a hit, and the ball outran the detection hardware: past about two screen widths a second a ball moves further than its own diameter during the hundred milliseconds the laser is lit, and where it was when the trigger broke stops being a meaningful question.
- **The ball never shrinks** in this drill. A smaller ball was reported unhittable, and a target smaller than the laser's own dot is a test of the detector rather than of marksmanship.
- **A new ball never appears on top of the muzzle.** It comes back at the far side of the playfield from your last hit, so nobody wins a level by standing still.

Missing costs nothing but time. Punishing a miss teaches flinching, and the clock punishes it already.

3 - Bouncing Balls



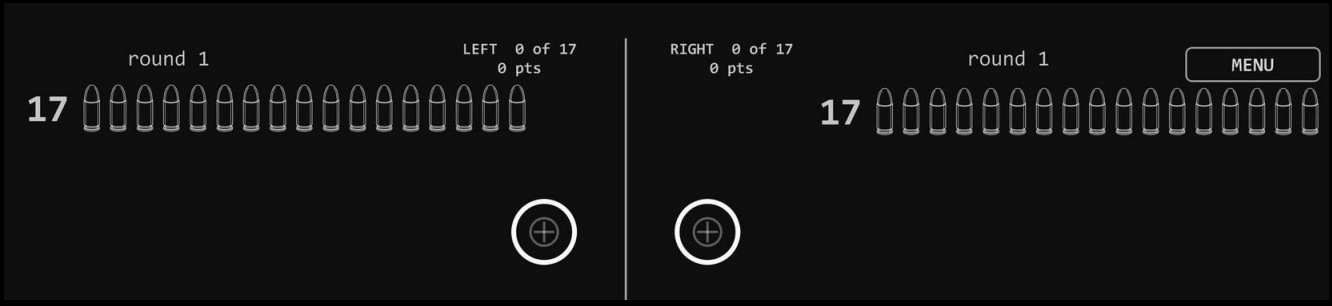
Several balls, each carrying its own level.

Several balls at once, and **each one carries its own level**, printed on it. Hitting the easy one does not punish you on the hard one, so after a few minutes the screen holds a spread of difficulties at the same time: the ball you keep hitting becomes the one you can barely hit, while the others wait. That is what makes it a drill rather than a single escalating target.

Five levels again, each five per cent faster and five per cent smaller than the last. The shrink is half what it was originally, because the complaint was precisely about small targets; and there is a floor on the radius, set from the measured size of the laser dot on real hardware.

Both ball drills draw a small target face on the ball rather than leaving it a plain disc. The face is drawn at a little over half the brightness ceiling and in thin strokes, because every unit of brightness under the dot comes off that shot's score: an earlier solid fill cost about fifty-five units and took a typical shot from 85 down to 30. The target was hiding the shots that hit it.

4 - Reaction



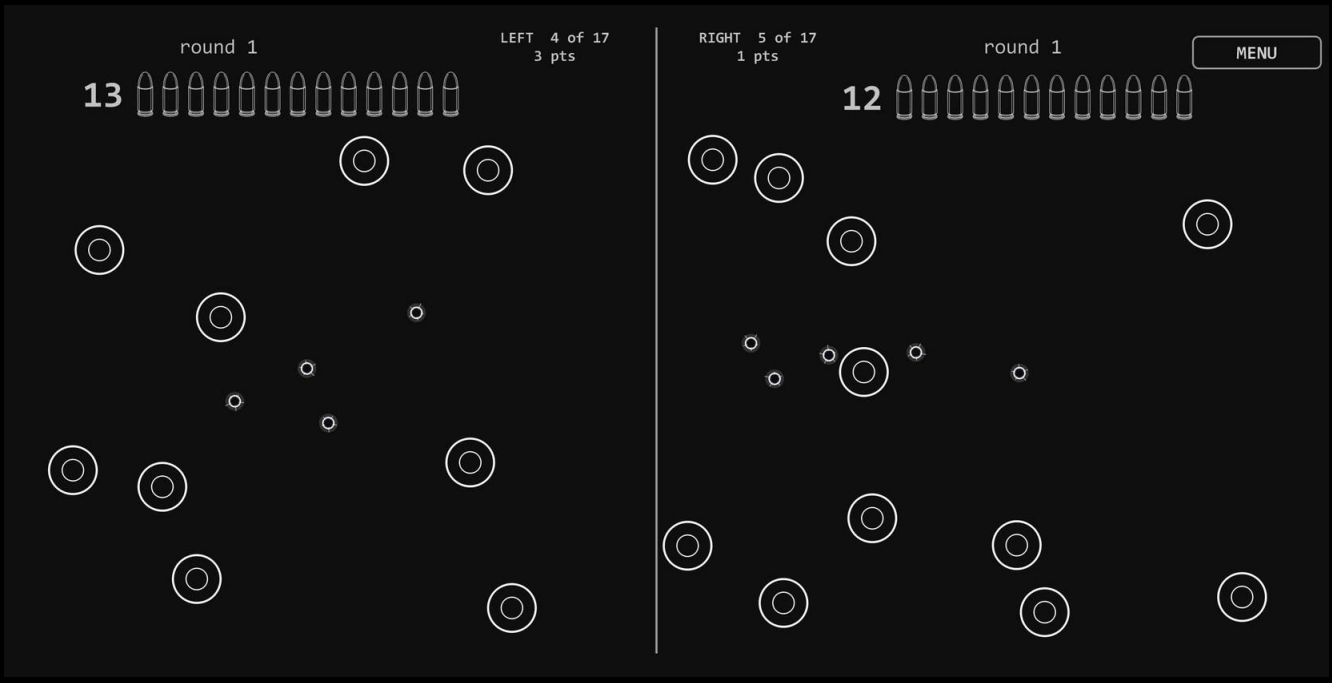
One target, revealed without warning. In two-player mode both sides see it at the same instant.

A blank screen, a wait of unpredictable length, and then one target. The clock starts the instant it is drawn and stops when you hit it.

- **The delay is random**, between 1.2 and 3.5 seconds. A fixed delay is trained against within a few repetitions, and then it measures your counting rather than your reaction.
- **Firing before the target appears is a false start.** It is reported and never scored, even if the target arrives a moment later. A shot that left early was a shot at nothing, and counting it would reward the one habit the drill exists to break.
- **The target moves** at least a third of the screen between repetitions, so it cannot be pre-aimed during the wait.

It reports your best and your **median** time. The median is the honest one: a single lucky repetition is not a skill.

5 - Clear The Board



Twelve circles, one clock.

Twelve targets, one clock, hit them all. It reports the time and the number of rounds it took.

Targets never overlap and never touch an edge, because an ambiguous hit between two circles or a target half off the screen makes the result arguable in exactly the wrong direction. A hit is scored against the circle it landed on and not the nearest circle: shooting between two targets is a miss, and generosity there would quietly inflate every hit rate the app reports.

When the board is cleared it pauses for a couple of seconds on your time and then deals a fresh one, so the screen is never just empty while you wonder whether something broke.

THE DRILLS: SIX TO NINE, THE FIGURES

Four drills share one cast of fifteen hand-drawn figures. They differ in what the figure does, not in how it is scored.



One of the fifteen. The scoring rings are printed in the drawing itself.

What they all have in common

The rings that score are the rings you can see. They are printed in the artwork, and the app reads them out of the ink as polygons rather than guessing a shape over the top. An early attempt fitted ellipses and rounded rectangles to them; the printed rings are bottle-shaped and no analytic shape followed them, so the drawing was taken as the authority instead. What you are scored against and what you are aiming at are the same line.

A figure whose rings could not be verified is never drawn at all. Better a smaller cast than a figure scored wrongly.

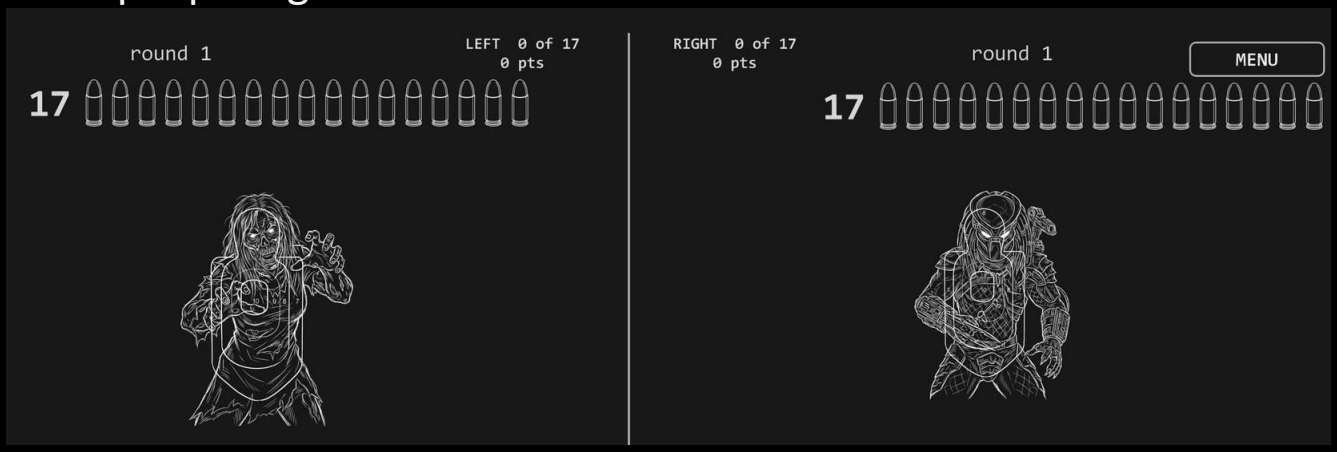
The cast is variety, not information. Nothing is scored on which figure appears. The next one is drawn at random, never the one just shown, and placed well away from where the last one stood.

A hit on the figure but outside every ring still shows. It leaves a dim grey mark and scores nothing. For a long time such a shot drew nothing at all, which from the firing line reads as “shots land inside the target and nothing happens”. Note that the head carries a printed 9 and a 7 with no ring behind them: a hit there is on the figure, not in a scoring zone.

Holes ride the figure. A shot that lands on a figure becomes part of it, stored in the figure’s own coordinates and redrawn as it moves and scales, so a walking target carries its hits with it and takes them away when it leaves.

Thirty points drops it. Each scoring hit adds its ring value to a running total shown beside the figure. At thirty the figure goes down into its own fall pose for a second, and the next one comes. The total belongs to that figure; the next one starts from zero.

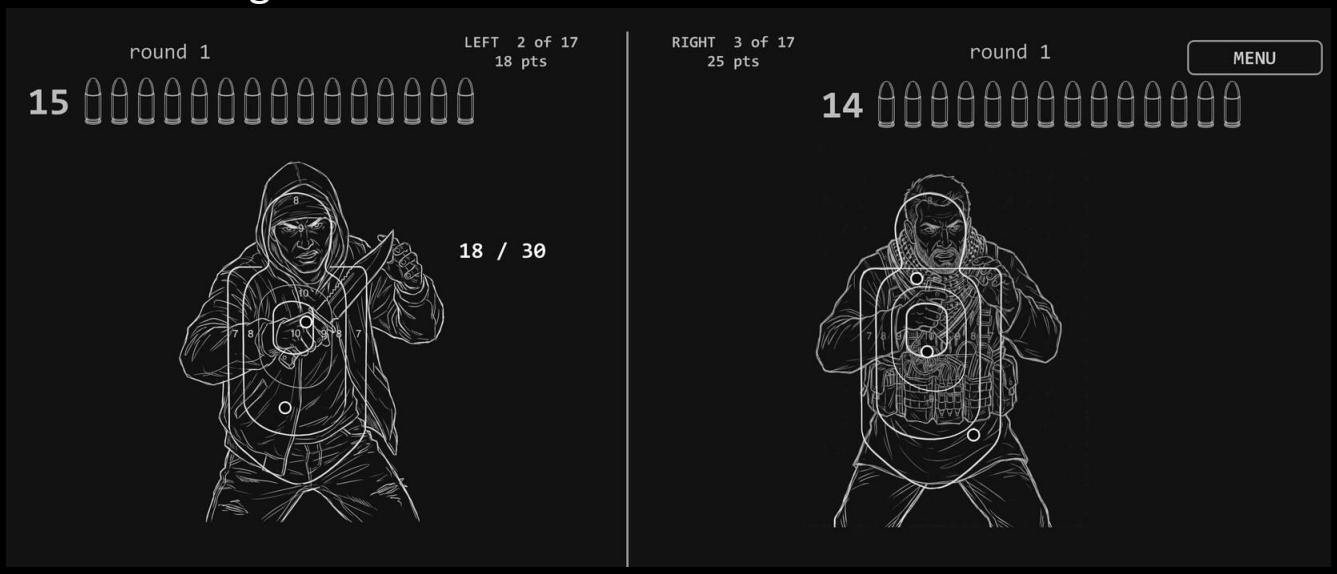
6 - Pop-Up Target



Two different figures, revealed at the same instant, one per side.

The reaction drill with a figure as the target: a blank screen, an unpredictable wait, then a figure at an unpredictable place and an unpredictable size, which stays for a few seconds and leaves whether or not you hit it. Firing before it appears is a false start and is never scored. It measures the time from the figure appearing to your first hit, the rings those first hits found, and the figures that left unengaged.

7 - Static Target

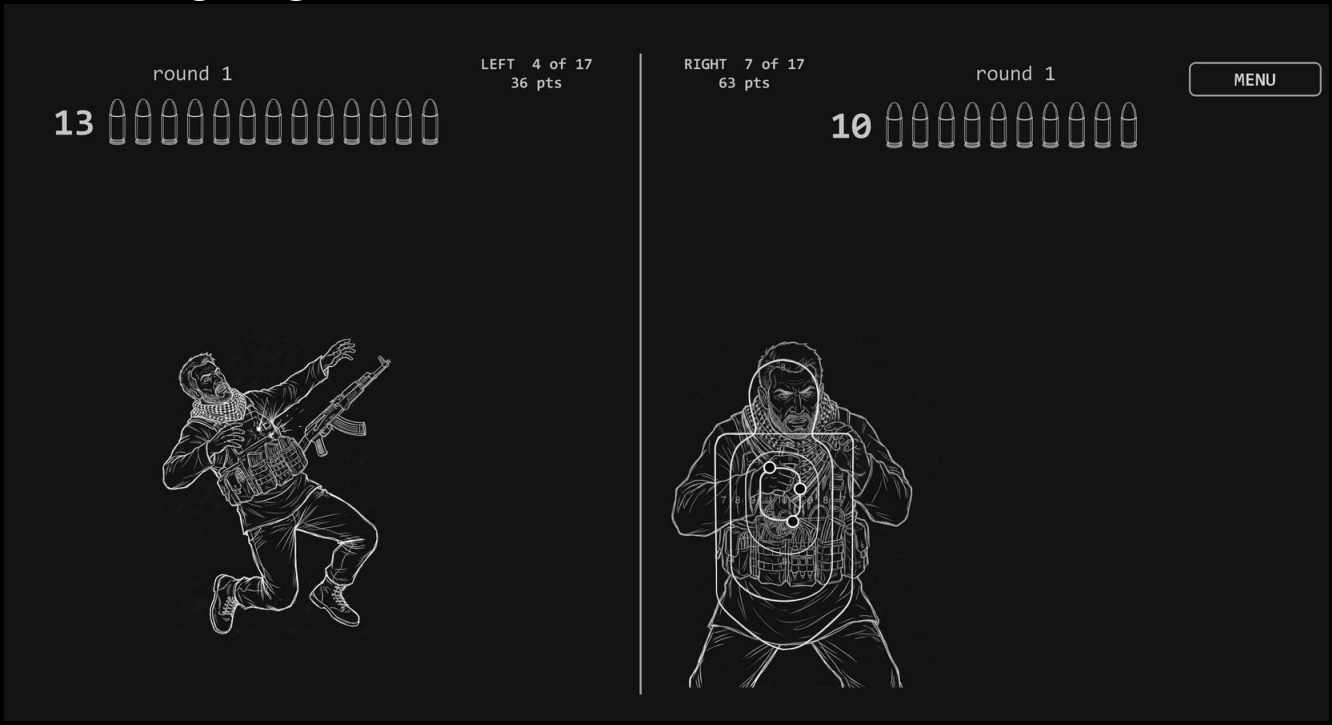


One figure, held, with its running total beside it.

One figure, standing, until you drop it. The counter beside it shows the running total rather than the value of the last hit – *"I want you to flash the total number, not just the number I hit"* – and the holes stay on the figure while it stands. A new round is a new figure, and it takes its holes with it.

This is the paper-target readout: where the shots went, what they scored, and how long the string took.

8 - Moving Target



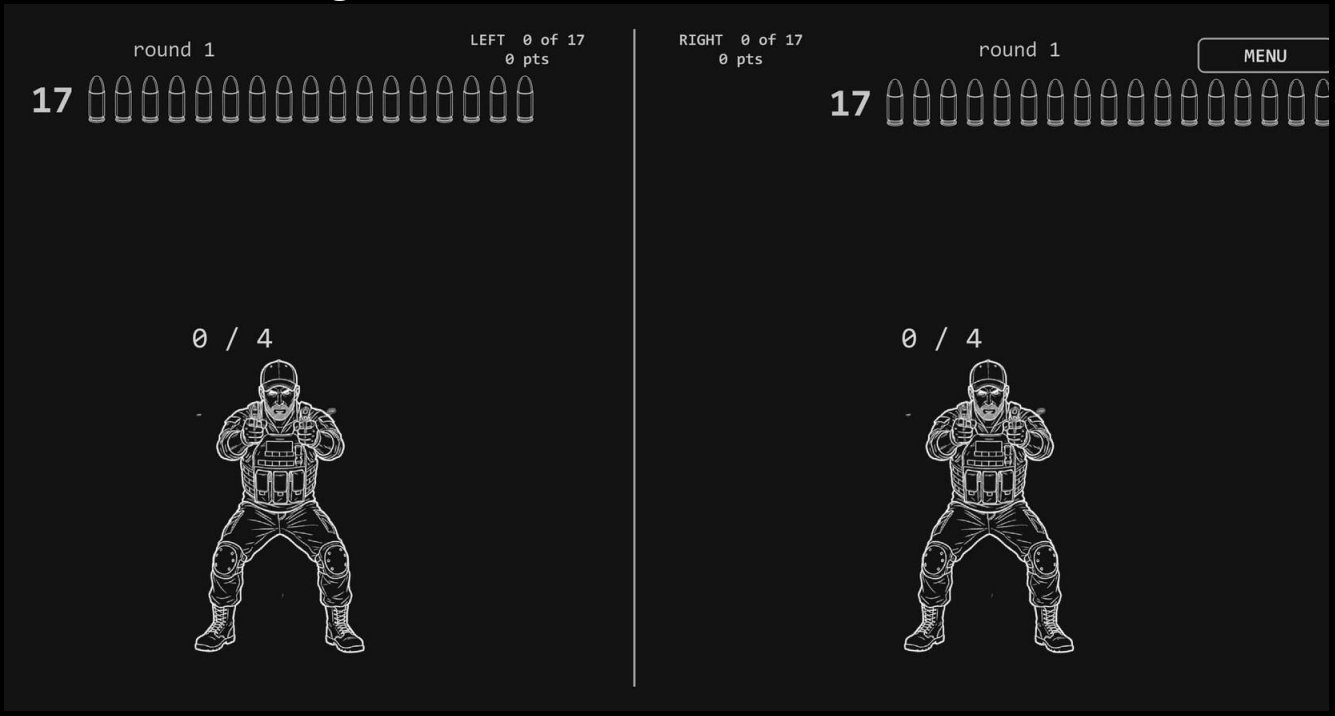
On the right, a figure holding at a distance. On the left, one that has just been dropped.

A figure walks toward you, holds, and retreats, drifting sideways as it goes. **The size is the distance**: it grows as it approaches and shrinks as it goes away. Everything about a cycle is drawn at random – which figure, how long the wait, which side it comes from, how far away it starts, how long it approaches, holds and retreats.

It reports hit rate and score in three distance bands, and the time from each hold beginning to the first hit.

One detail that matters more than it sounds: a shot is confirmed two camera frames after the trigger breaks, by which time a walking figure has moved and grown. The hit test therefore rewinds to where the figure was *and how big it was* at the moment of the shot. Without that rewind you would have to shoot in front of the target, which is exactly what the ball drills used to require before the same fix was applied to them.

9 - Animated Target



The filmed soldier, and the hit counter above him.

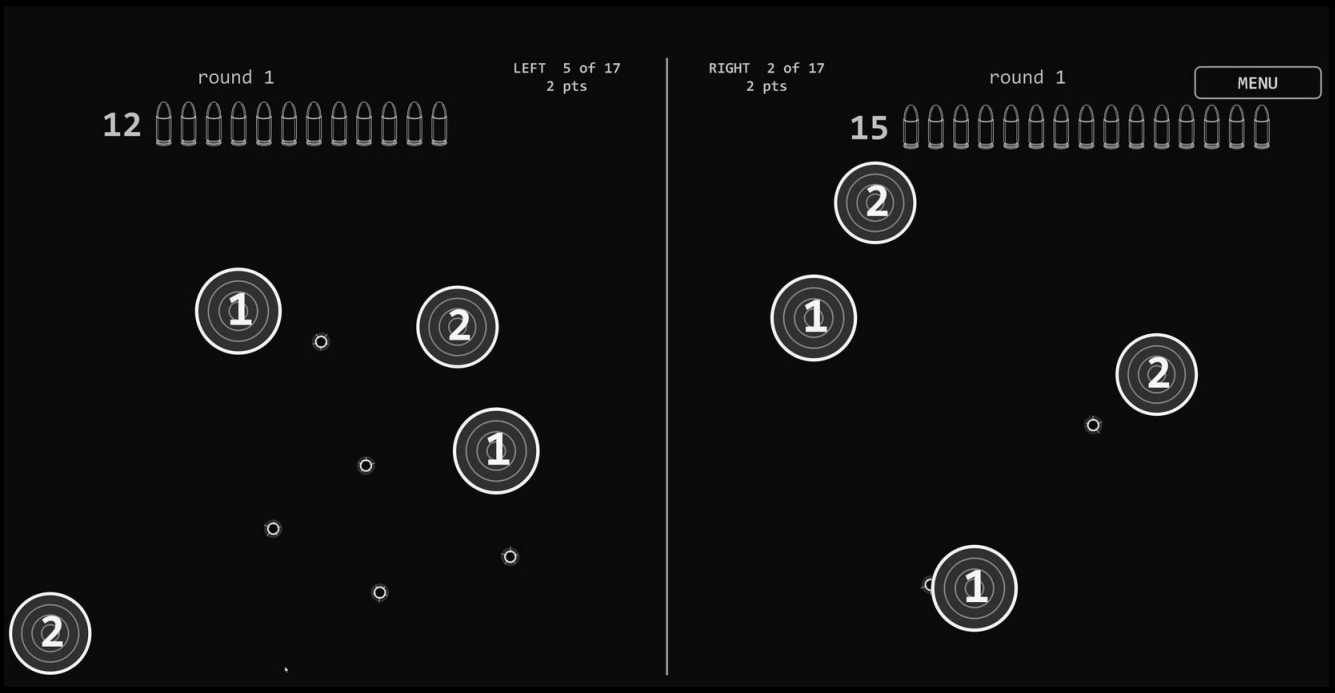
A filmed figure, ten seconds of it, moving continuously – aiming, firing, crouching and rising. There are no rings printed on him, so he is scored the only honest way left: a shot inside his silhouette scores, a shot outside it does not, and the silhouette is the shape you can see rather than a box around it.

Four hits in the head or centre mass drop him. A hit anywhere else on him scores ten points but does not advance the counter, and the counter above him says *HIT - NO COUNT* for a moment so you know why. He is never marked with holes: he moves every frame, and holes on him would be new lit area on every one of them. The counter carries the feedback instead.

When he goes down he plays his own fall, rests for a second, and comes back.

TWO PLAYERS

Press **T**, or shoot the Two Players tile. The screen splits down the middle and two people shoot at once, one per side, at the same drill. **O** puts it back to one. Every drill in the app runs two-player.



Two players, one drill, one magazine each.

Why the split is left and right

Type sizes, the reserved band at the top and the strip at the bottom are all worked out from the height of the screen. Splitting left and right keeps every one of them exactly as it is in single player. A top-and-bottom split would have halved the height and shrunk every piece of text on a screen that is read from several metres away.

Each half gets exactly half of what is left after the divider, so the two playfields are equal by construction. A shot to the left of the divider's centre line belongs to the left player and anything from the centre line rightwards belongs to the right: one comparison, no dead zone, no argument.

A round is one magazine each

Each side has its own strip of seventeen and its own score. A side is finished when the app has seen and scored seventeen of its shots.

That is a deliberate choice about fairness. If the app under-detects one shooter's pistol, that shooter simply pulls the trigger a few more times to reach seventeen registered shots. Both sides therefore always reach exactly the same denominator, the round always closes by itself, and no guess about shots the camera never saw ever enters the result.

"If the shots are missed, they are missed randomly from both sides. Just count the shots that had been registered."

One reveal, shared

In the Reaction drill both halves are shown their target at the same instant. Two independent copies would each draw their own wait, one player would see his target up to two seconds before the other, and the drill would measure luck rather than reaction.

A reveal that one side never resolves does not hang the other: after a timeout that side is marked **TOO SLOW** – counted, never a hit and never a false start – and the next reveal goes up for both.

Different figures, same instant

In the figure drills each side gets a *different* figure, appearing at the same moment, mirrored so the pair is symmetric about the divider. The cost of that is known and was accepted: the figures are not all the same size, and the ten-ring area varies by about 1.6 times across the cast. It is a game between two people in a room, and watching your opponent shoot a different villain is worth more than the last few per cent of fairness.

Who wins

Points win the round. Everything else is shown beside the result and may disagree with it:

LEFT *RIGHT*
17 of 17 shots *17 of 17 shots*
165 of 170 points BEST *120 of 170 points*
60.0 seconds *10.0 seconds*
EFFICIENCY 2.75 *EFFICIENCY 12.00 BEST*

SCORE 65 *SCORE 282*
AIM 97% SPEED 7% *AIM 71% SPEED 40%*
BEST SCORE SO FAR 212 *BEST SCORE SO FAR 212*
BEST EFFICIENCY SO FAR 3.99 *BEST EFFICIENCY SO FAR 12.00*

ROUND 2 COMPLETED
LEFT WINS by 45 points
RIGHT is more efficient, 12.00 to 2.75

[NEXT ROUND]

That example is the specification, not an accident. A shooter who takes a minute to place careful hits wins on points, while a shooter who empties the magazine in ten seconds wins on efficiency – and, here, on the score as well, because the score counts speed and the round does not. Both sides' figures are shown and the better of each is marked, so the trade is visible rather than inferred, and the argument afterwards is about which number you care about rather than about what happened.

Each side's record book is the same one described in *The Score*: a two-player half is stored separately from a solo round, because it is a different test, but there is no shooter identity in it. The two of you share one book.

The sides swap every round and the totals follow the player rather than the half of the screen, so a better view, a better angle or a brighter corner cannot decide a match. The swap is not announced on screen; it is in the log.

WHY THIS IS FAMILY FUN

Most dry-fire practice is solitary and silent. This is the opposite, and nearly every decision in the app points the same way.

Two people, one screen, no equipment to divide. Press T and the screen splits. There is nothing to pair, nothing to log in to and nothing to configure per player. Whoever is holding the second pistol is in the game.

A round is short. Seventeen rounds takes twenty or thirty seconds. Nobody waits long for a turn, and a whole evening is a stack of rounds rather than one long session.

The numbers show their working. The efficiency has its arithmetic printed underneath it and the score shows both the factors it is made of, so nobody has to take the app's word for anything. Points win the round, the careful shooter and the fast shooter each get a column to point at, and what is left to argue about is which number matters – which is the good kind of argument.

The sides swap every round. The result cannot be blamed on which half of the screen anybody had.

The drills explain themselves. A ball you have to hit, targets you have to clear, a figure that pops up. Nobody has to read this manual to play, which is the only honest test of a game.

The targets are drawings, not abstractions. Fifteen hand-drawn figures, each with scoring rings printed on it, plus a filmed soldier who takes four good hits and goes down. A child who is bored by a circle is not bored by a zombie.

It is entirely offline. No account, no network, no subscription, nothing installed, nothing phoned home. Unzip a folder and shoot.

What you hear

Every registered shot plays a gunshot. That bang is the app's way of saying it saw the trigger pull, which is why nothing is ever played *instead* of it: an effect follows a millisecond later and never replaces it.

Sound	When
Gunshot	Any registered shot
Bubble pop	A ring target is hit, and a pull on an empty magazine
Glass breaking	A ball bursts
Notification	A target appears in Reaction
Shrill horn	A figure appears
A grunt, or a banshee	A figure goes down

The app also speaks: it announces each drill by name, warns you to hold fire while it is measuring, and tells you when it believes the camera has moved. The lines are recorded in three different voices and one is chosen at random, so the app does not sound like one announcer reading a script. Nothing is spoken within three seconds of a trigger pull, except the warnings that must be heard mid-string.

+ and **-** set the volume, and the setting is remembered. The floor is one per cent rather than zero: a key that could silence the app entirely would bring back a fault that took three days to find, when every shot was being played correctly into a volume slider somebody had dragged to the bottom.

EVERY KEY

Press **K** at any time and the app shows this screen itself.



The Keyboard Shortcuts screen, as the app draws it.

The drill rows on that screen are read from the app's own list of drills, using each drill's own name, so a drill added in a later version appears there by itself and a renamed one cannot be mis-described. It exists because the key list it replaced was typed by hand and spent three days advertising a drill that had been removed.

Key	Drill
1	Static Targets
2	Escalating Ball
3	Bouncing Balls
4	Reaction
5	Clear The Board
6	Pop-Up Target
7	Static Target
8	Moving Target
9	Animated Target
M	Main Menu
K	Keyboard Shortcuts
N	New round

Pressing a drill's own key again starts a new round of it.

Key	On the menu
O	One player
T	Two players

Key	Checks
C	Self-calibrate
A	Self-check
Q	Show the last verdict
G	Calibrate by shooting a grid
X	Measure the exposure
F	Measure the noise floor

Key	Show or hide
0	All the readouts at once
S	Stats
B	Status band
L	Signal meter
P	Camera pane
V	Aim the camera

Key	Also
E	Erase the holes
+ -	Volume up and down
Esc	Exit the game

The shortcuts screen claims no keys of its own. From it, pressing 1 starts drill 1 and C calibrates, exactly as if you were on a drill. A shot fired at it draws its hole and scores nothing, so the hit rate is not deflated by shots at a help screen.

LOOKING AFTER IT

Closing it properly

Escape, or the **Exit** tile shot twice, or *stop_screenplink.ps1*.

Do not End Task it and do not force-kill it. Cutting a streaming USB camera off mid-transaction can hang the camera's own firmware, and then the only cure is to unplug it and plug it back in. On one occasion it took down the whole USB hub and both cameras with it. Every graceful exit route in the app exists because of that.

Running two copies at once does the same thing: the older one loses the camera mid-stream. The launcher refuses a second instance for that reason.

When something changes

What changed	What to do
You moved or knocked the camera	**C**, or **G** if the markers will not confirm
The light in the room changed	**X**
Something bright was on screen during setup	**F**
The screen is cluttered	**0**, then **E** to wipe the holes
You are not sure it is seeing your shots	**A**, then **Q** to re-read the verdict

A mapping measured in one camera position is valid only in that position, and nothing can detect after the fact that a camera has been nudged. If holes start landing away from the shots, that is the first thing to suspect.

Where your things are kept

Everything lives beside the executable, so the whole folder can be copied to another machine:

config\settings.local.json *what has been measured on this machine, and the best round it has seen*
config\calibration.json *the camera-to-screen mapping, and the sensitivity it has learned from real shots*
config\settings.default.json *what the app assumes before measuring anything*
logs *an account of every launch*

To start completely fresh, delete *settings.local.json* and *calibration.json*. The app measures everything again on the next launch.

To clear the **records** only, delete the *bests* section inside *settings.local.json* and leave the rest of it alone. Every best score and best efficiency goes; nothing the app measured about your camera or your room does.

One thing worth knowing before you delete them. The mapping rebuilds itself in about ninety seconds, but the sensitivity map – which evens out the fact that the far side of an angled screen is dimmer to the camera than the near side – is learned only from real shots over time. A folder that has been shot in for a week is genuinely better than a fresh one on the same hardware in the same room. Back those two files up rather than throwing them away.

If it is not working

Doctor.cmd tests every camera on the machine and writes *doctor-report.txt*. Run it with the app closed: two programs on one camera can leave the driver in a bad state.

The folder also contains a *CLAUDE.md* written for Claude Code, and a *.mcp.json* that lets it query the running app directly for what the camera is seeing and what the detector measured. Opening the folder in Claude Code and asking it to set the app up is the fastest route through a difficult camera.

Three diagnoses that are almost never what they look like:

- **Shots register when you stand square to the screen and stop registering when you move to one side.** That is the screen, not the app. A glossy or anti-reflective panel sends the beam away from the camera when you shoot at an angle, and nothing can be adjusted to recover it. See *The Screen It Needs*.
- **Phantom shots** are something coloured on the monitor the camera watches, not a fault in the detector. Move other windows off that screen.
- **A weak signal is not a reason to darken the room.** A dark room is the best condition this app can have. The cures are a longer exposure (**X**), a camera closer to the screen, or a shorter range.